

Projective polynomials and some of their applications

Lukas Kölsch

Department of Mathematics and Statistics,
University of South Florida

June 2026

Abstract

Projective polynomials over a field of characteristic $p > 0$ are polynomials of the form $x^{[k]} + c_{k-1}x^{[k-1]} + \cdots + c_1x^{[1]} + c_0x^{[0]}$, where $[i] = (q^i - 1)/(q - 1)$ for some $q = p^n$. I shall recall some classical results on these polynomials, connecting them to linearized polynomials as well as skew polynomials. I will then survey several constructions and results in combinatorics, geometry, and cryptography that make use of projective polynomials. In particular, we will focus on highly nonlinear functions. Here, the construction of many of the known infinite families with good properties can be framed through the lens of projective polynomials. This leads in particular to a unified framework, simpler proofs, and the discovery of new families.

This talk is based on joint work with Faruk Göloğlu.